Circuite în regim nesinusoidal

$$M(t) = M(0) + \sum_{k=1}^{\infty} M^{(k)} \cdot \sqrt{2} \sin(k\omega t + \varphi^{(k)})$$

k = ordin armonică

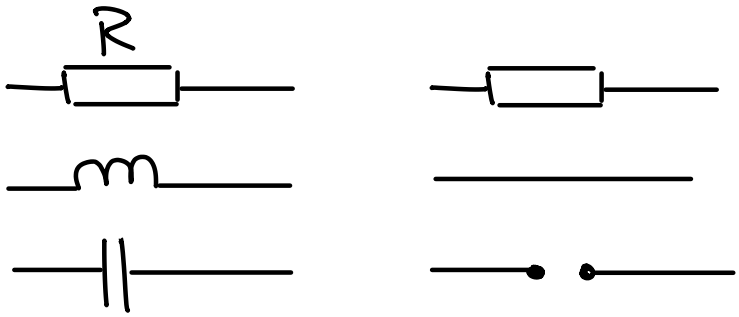
$M(0)$ = componentă curent continuu

ex $j(t) = 2 + 5 \sin \omega t + 2\sqrt{2} \sin 3\omega t$

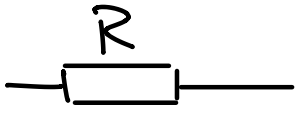
- 2 → c.c. $k=0$
- $5 \sin \omega t$ → c.a. $k=1$
- $2\sqrt{2} \sin 3\omega t$ → c.a. $k=3$

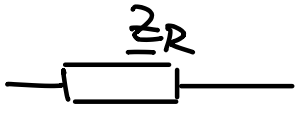
Elemente în regim nesinusoidal

→ în continuu

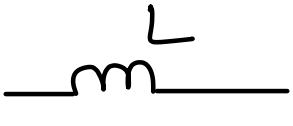


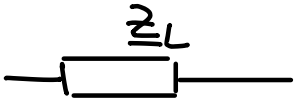
→ în alternativ



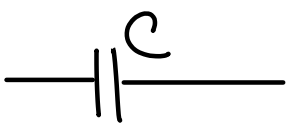


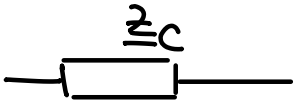
$\underline{Z}_R = R$





$\underline{Z}_L = k j \omega L$





$\underline{Z}_C = \frac{-j}{k\omega C}$

Puteri în regim nesinusoidal

$$u(t) = U^{(0)} + \sum_{k=1}^{\infty} U^{(k)} \sqrt{2} \sin(k\omega t + \alpha^{(k)})$$

$$i(t) = I^{(0)} + \sum_{k=1}^{\infty} I^{(k)} \sqrt{2} \sin(k\omega t + \beta^{(k)})$$

Puterea activă

$$P = U^{(0)} I^{(0)} + \sum_{k=1}^{\infty} U^{(k)} I^{(k)} \cdot \cos \varphi^{(k)} \quad [W]$$

$$\varphi^{(k)} = \alpha^{(k)} - \beta^{(k)}$$

Puterea reactivă

$$Q = \sum_{k=1}^{\infty} U^{(k)} I^{(k)} \sin \varphi^{(k)} \quad [VAR]$$

Puterea aparentă

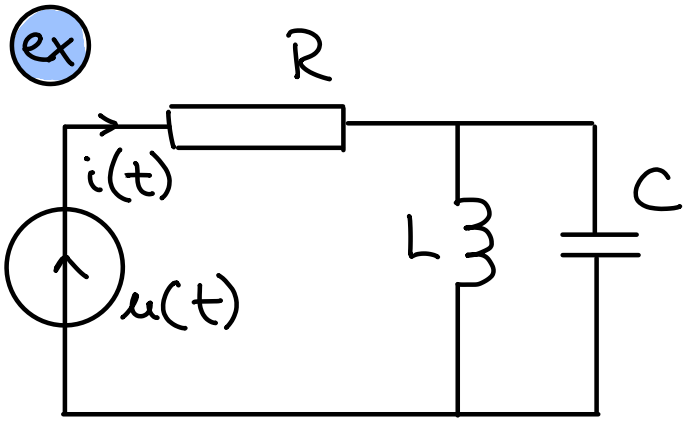
$$S = U \cdot I = \sqrt{U^{(0)2} + U^{(1)2} + U^{(2)2} + \dots} \cdot \sqrt{I^{(0)2} + I^{(1)2} + I^{(2)2} + \dots}$$

$$[VA]$$

Puterea deformantă

$$D = \sqrt{S^2 - P^2 - Q^2} \quad [VAD]$$

ex



$R = 6\,\Omega$

$L = 0{,}2\,H$

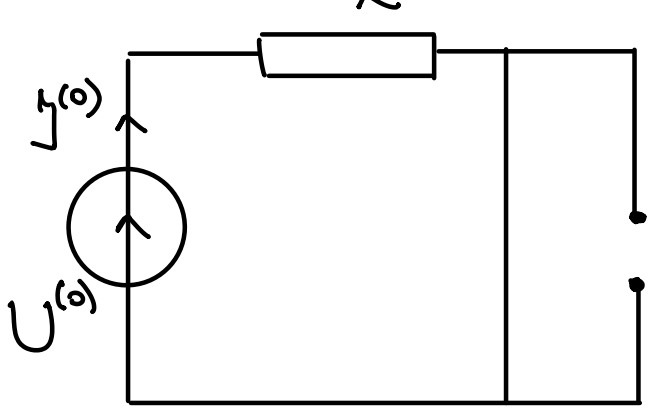
$\omega = 10\ rad/s$

$C = \frac{1}{90}\,F$

$u(t) = 12 + 24\sqrt{2}\sin(3\omega t + \pi/2)$

$i(t) = ?$

→ componenta c.c. ($k=0$)

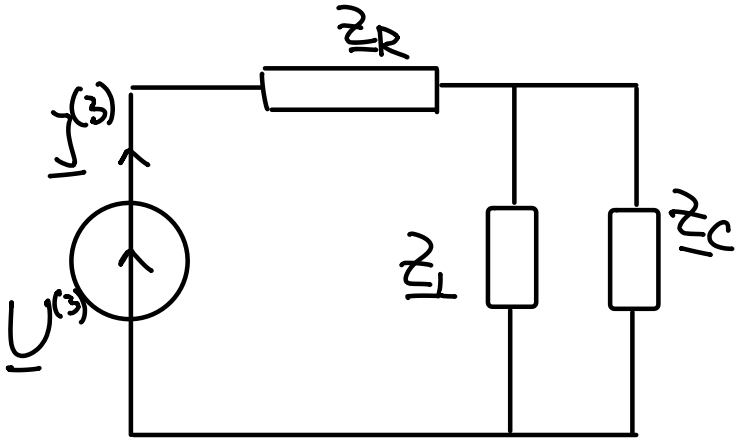


$U^{(0)} = 12\ V$

$y^{(0)} = \frac{U^{(0)}}{R} = \frac{12}{6} = 2A$

→ componenta c.a. ($k=3$)

$u^{(3)}(t) = 24\sqrt{2}\sin(3\omega t + \pi/2)$



$\underline{Z}_R = R = 6$

$\underline{Z}_C = \frac{-j}{k\omega C} = \frac{-j}{3 \cdot 10 \cdot \frac{1}{90}} = -3j$

$\underline{Z}_L = jk\omega L = 3 \cdot 10 \cdot 0{,}2j = 6j$

$$\underline{U}^{(3)} = 24 e^{j\pi/2} = 24\cdot(\cos \pi/2 + j \sin \pi/2) = 24(0+j) = 24j$$

$$\underline{Z}_{ech} = \underline{Z}_R + \frac{\underline{Z}_C \cdot \underline{Z}_L}{\underline{Z}_C + \underline{Z}_L} = 6 + \frac{-3\cdot 6j^2}{-3j+6j} = 6 + \frac{18}{3j} = 6-6j$$

$$\underline{Y}^{(3)} = \frac{24j}{6-6j} = \frac{4j}{1-j} = \frac{4j(1+j)}{2} = -2+2j$$

$$i^{(3)}(t) = 2\sqrt{2}\cdot\sqrt{2}\sin(3\omega t + 3\pi/4)$$

$$i(t) = \underline{Y}^{(0)} + i^{(3)}(t) = 2 + 4\sin(3\omega t + 3\pi/4)$$

Puter. activ. $P = U^{(0)}Y^{(0)} + U^{(3)}Y^{(3)}\cos\varphi^{(3)}$

$$P = 12\cdot 2 + 24\cdot 2\sqrt{2}\cos\left(\frac{\pi}{2} - \frac{3\pi}{4}\right)$$

$$P = 24 + 48\sqrt{2}\cos\left(\frac{-\pi}{4}\right) = 24 + 48\cdot\sqrt{2}\cdot\frac{1}{\sqrt{2}}$$

$$P = 72\ W$$

Puterea react. $Q = U^{(3)}Y^{(3)}\sin\varphi^{(3)}$

$$Q = 24\cdot 2\sqrt{2}\cdot\sin\left(\frac{\pi}{2} - \frac{3\pi}{4}\right)$$

$$Q = 48\sqrt{2}\cdot\sin\left(-\frac{\pi}{4}\right) = 48\cdot\sqrt{2}\left(-\frac{1}{\sqrt{2}}\right)$$

$$Q = -48\ VAR$$

Puterea aparentă

$$S = U\cdot Y = \sqrt{U^{(0)2} + U^{(3)2}}\cdot\sqrt{Y^{(0)2} + Y^{(3)2}}$$

$$S = \sqrt{144+576}\cdot\sqrt{4+8} =$$

$$S = \sqrt{720}\cdot\sqrt{12} = 3\cdot 4\sqrt{5}\cdot 2\sqrt{3}$$

$$S = 24\sqrt{15} \text{ VA}$$

Puterea deformanta

$$D = \sqrt{S^2 - P^2 - Q^2}$$

$$D = \sqrt{8460 - 5184 - 2304}$$

$$D = \sqrt{972} = 2 \cdot 9\sqrt{3} = 18\sqrt{3} \text{ VAD}$$